one, because the gravitational waves are disturbed less by matter than the electromagnetic ones and were able to travel undisturbed even when the universe was too dense to let the electromagnetic waves pass.

We have now observed gravitational waves directly, with the LIGO detector, formed by two arms of a few miles in length, at a right angle to each other, in which laser beams measure the distance between three fixed points. When a gravitational wave passes, the space lengthens and shortens imperceptibly, and the lasers reveal this minuscule variation.[fn46] The gravitational waves observed were generated by an astrophysical event: colliding black holes. These are phenomena described by general relativity which do not involve quantum gravity. But a more ambitious experiment called LISA is at the stage of being evaluated and is capable of doing the same thing but on a much larger scale: by putting into orbit three satellites, not around the Earth but around the Sun, as if they were miniature planets tracking the Earth in its orbit. The three satellites are connected by laser beams measuring the distance between them or, better still, the variations in the distances when a gravitational wave passes. If LISA is launched, it should be able to see not only the gravitational waves produced by stars and black holes but also the diffuse background of primordial gravitational waves generated at a time close to the Big Bang. These waves should tell us about the quantum bounce.

In the subtle irregularities of space, we should be able to find traces of events which took place 14 billion years ago, at the origin of our universe, and confirm our deductions on the nature of space and time.